

JACOB COLVIN

Lead Software Engineer; Developer Platforms & Infrastructure

📍 Cincinnati, Ohio

✉ me@jacobcolvin.com

🌐 jacobcolvin.com

📱 colvinjm

🔌 MacroPower

ABOUT

I am a **Lead Software Engineer** with 10 years of experience in **developer platforms** and **infrastructure**, and deep roots in **Site Reliability Engineering**. I build robust developer platforms on **Kubernetes**, craft maintainable **infrastructure as code**, and architect systems that are simple and straightforward to operate. My experience as a developer, with a focus on **Go** and **Python**, lets me create platforms that are not only reliable, but also enjoyable to use.

EXPERIENCE

84.51° / The Kroger Company

📍 Remote / Cincinnati, Ohio

📅 May 2016 – Current

Lead Software Engineer (Platform Engineering)

📅 March 2025 – Current

- Set technical direction for our cloud engineering team as an IC lead, shaping architectural decisions across **Kubernetes (AKS)**, enterprise-wide **IaC** patterns, and observability platforms.
- Led migration of **JFrog Artifactory** onto **Kubernetes (AKS)**, with infrastructure and configuration fully managed via **Terraform** and **Flux**; quadrupled maximum capacity and eliminated a recurring stream of major incidents.
- Cut Azure CosmosDB logging costs by over 90% (\$236k/yr in realized savings) with a **Vector** log pipeline.
- Established a repeatable process for **Azure** compute reservations, saving \$56k/yr on the initial purchase.
- Drove decommissioning of **VMware Tanzu Application Service (TAS)**, owning planning, company-wide communication, and migration support; retired 500+ VMs and enabled renewal of VMware/Broadcom contracts at a lower rate.
- Designed a new **Terraform**-based management system for multi-region **Kubernetes (AKS)**, incorporating **Cilium**, **Karpenter**, **Azure Service Operator**, and **Kyverno**.
- Shipped six feature releases of a company-wide **Python** instrumentation library, expanding support to new runtimes including **Databricks Serverless** and **Azure Function Apps**.
- Acted as a go-to escalation point across the organization for **Kubernetes**, **Azure**, and observability platforms, unblocking partner teams through direct troubleshooting, clear documentation, and self-service guidance.
- Pioneered **AI**-assisted engineering workflows on the team, establishing tooling and repeatable practices for agentic development against **IaC** and **GitOps** repositories; substantially increased delivery speed and mentored teammates on best practices.

Senior Site Reliability Engineer (Kubernetes)

📅 January 2023 – February 2025

- Headed design and maintenance of our **IaC (Terraform, Flux)** for **Kubernetes (AKS)**, supporting new production workloads responsible for \$100MM+ in operating profit, while cutting manual toil by 200+ hours per year.
- Developed **Python** application for scaffolding template repositories, as well as centralized **Helm charts** and related **GitHub Actions**, which collectively saved developers 3000+ hours during the first year in production.
- Led **Datadog** cost-saving efforts which reduced spend by over \$400k per year; enforced controls using **Terraform**.
- Architected an **Azure EventHub / Vector** pipeline which cut log ingestion and indexing costs by 85%, enabling migration from **Log Analytics** to **Datadog** with over \$650k in net savings during its first year.
- Reduced **Kubernetes (AKS)** compute costs by roughly 30% through automated cleanup policies, right-sizing of system workloads, and improved developer defaults.
- Replaced per-deployment secret-sync sidecars with centralized **External Secrets Operator** controllers backed by **Azure Key Vault**, reducing cluster overhead and automating secret management.
- Drove adoption of **Datadog APM** via self-service **Helm** configuration and auto-instrumentation, onboarding 100+ applications and 400+ synthetic tests.
- Reworked **GitOps (Flux)** repositories to prioritize velocity and self-service, cutting validation cycle times from minutes to seconds; automated dependency updates across them with **Renovate**.
- Enabled partner teams to manage their own infrastructure independently, reducing our team's support load from 20+ hours per week to nearly zero.
- Acted as lead subject-matter expert for observability platforms (**Datadog**, **Grafana**, **Prometheus**, **Thanos**, **OpenTelemetry**, **Vector**), enabling developers to troubleshoot their applications independently.

Senior Site Reliability Engineer (Observability)

📅 April 2021 – December 2022

- Deployed and supported enterprise observability services/tooling, utilizing **Ansible** (on-premises) and **Flux** (Azure Kubernetes), including **Grafana**, **Prometheus**, **Thanos**, **Fluentd**, **Telegraf**, and **Jaeger**.
- Developed a **Python** library to centralize instrumentation for **Prometheus** metrics, **OpenTelemetry** tracing, structured logging, and **pprof** profiling, with support for **FastAPI**, **Databricks** notebooks, and more. It delivered a consistent experience across our observability platforms and directly saved developers a collective 1000+ hours.
- Led development of multiple **Prometheus exporters** using **Go**, utilizing **Redis** for caching, distributed workers with pub/sub and leader election, **CEL** for custom rule evaluation, **Cue** and custom tooling for validation, and Go text templating.
- Rolled out **Datadog** as a unified logging platform for both **Azure** and on-premises systems, using **Terraform** for configuration and **Fluentd** for log parsing and forwarding; enabled deprecation of multiple disparate logging solutions.
- Created multiple **Jsonnet** libraries for **Prometheus** and **Alertmanager** configuration, saving 100+ hours per year.
- Upstream contributor to **Grafana**, **Thanos**, **Jaeger**, and other open-source observability projects.

Versatilist Engineer (Cloud Infrastructure)

📅 January 2020 – March 2021

- Created and administered multiple **Azure** environments using **Terraform** in Azure DevOps; these environments frequently included resources such as **Databricks** workspaces, **Data Factory** instances, **Key Vaults**, **Storage Accounts**, and **Postgres** databases.
- Enhanced our **Prometheus** ecosystem by adding high-availability and long-term storage via **Thanos**, thus allowing consumers of Prometheus to track SLOs and KPIs over years instead of days.
- Assisted a mix of seven development and data science teams with **Azure** migration, **Terraform**, **Grafana** dashboards, **Prometheus** alerts, and **CI/CD** pipelines (**GitHub Actions**, **Azure DevOps**); acted as technical liaison to increase speed and accessibility of support.

ICT Co-op

📅 May 2016 – December 2019

- Designed a **C#** API layer over several legacy systems, and a SPA using **TypeScript** with **React**, to both assist with support and improve velocity towards deprecation of legacy systems.
- Created a **Prometheus exporter** for **SonarQube** data, along with corresponding dashboards, rules and alerts, to allow the security team to gather relevant KPIs.
- Automated **Red Hat Linux** VM deployment through **ServiceNow**, via interactions with **Satellite**, **vCenter**, **SolarWinds** and **Ansible**.
- Interfaced with **BOSH** and **Hadoop** to design custom **Pivotal Cloud Foundry** monitoring solution.

EDUCATION

B.S. Information Technology University of Cincinnati

📅 August 2015 – April 2020

- Summa Cum Laude
- Cybersecurity specialization

PROJECTS

kat ★ 164

🔗 MacroPower/kat

- TUI and rule-based rendering engine for Kubernetes manifests. Automatically invokes generators like Helm, Kustomize, or KCL, and provides a persistent, navigable view of rendered resources, with live reload, integrated validation, and diff visualization.
- Includes an MCP server that lets AI agents iterate on manifests through a consistent render/validate pipeline, without handing over cluster or command-line access.
- Technologies: Go, Bubble Tea, CEL

kclipper ★ 39

🔗 MacroPower/kclipper

- A superset of KCL (a CNCF configuration language) that integrates Helm: renders charts directly within KCL with full pre- and post-render control, declaratively manages charts and generated schemas, and runs as an ArgoCD Config Management Plugin.
- Technologies: Go, KCL, Helm

Homelab ★ 84

🔗 MacroPower/homelab

- Infrastructure-as-code for my homelab / personal cloud. Defines multiple interconnected Kubernetes clusters spanning bare metal (Talos) and Hetzner Cloud, orchestrated by ArgoCD ApplicationSets (100+ Apps, 50+ charts) defined in KCL via kclipper.
- Runs a fully self-hosted observability stack (Grafana, Mimir, Loki, Tempo, Alloy), Envoy Gateway for ingress, Crossplane, and a self-hosted Dagger CI engine.

- Technologies: Talos, Cilium, ArgoCD, KCL, Helm, Terraform, Envoy, Crossplane, Dagger

Analytics Panel Plugin ★ 30

MacroPower/macropower-analytics-panel

- Grafana panel plugin that injects JavaScript into dashboards, which reports user session information to a backend Go server, which in turn exports Prometheus metrics for display in (you guessed it) Grafana.
- Over 1MM downloads, featured by Giant Swarm:
<https://www.giantswarm.io/blog/grafana-ception-or-how-we-do-grafana-analytics-giant-swarm>
- Technologies: TypeScript, React, Go

Prometheus Video Renderer ★ 111

MacroPower/prometheus_video_renderer

- Just for fun, completely impractical tool that allows you to encode audio and video as Prometheus metrics.
- Featured on the Grafana blog:
<https://grafana.com/blog/2021/07/30/how-to-use-grafana-and-prometheus-to-rickroll-your-friends-or-enemies>
- Technologies: Go, Jsonnet

Wakatime Exporter ★ 123

MacroPower/wakatime_exporter

- Prometheus exporter and Grafana dashboards for Wakatime coding statistics.
- Over 100k downloads
- Technologies: Go